

Context

This activity aims to deepen understanding of the OSPF protocol using an Ethernet network topology where, unlike P2P connections, there are several routers connected to the same network segment. In these cases, to limit the amount of OSPF signaling that would be required to create adjacencies between all routers, OSPF messages are sent centrally to the Designated Router (DR) or its backup, the Backup Designated Router (BDR).

Criteria for DR and BDR Selection:

Router Priority:

- Each OSPF router has a configurable priority value for each network interface.
- The priority can range from 0 to 255. The router with the highest priority is selected as DR, and the second highest as BDR.
- If the priority is set to 0, the router is not eligible to be DR or BDR.

Router ID (RID):

- If two or more routers have the same priority, the tiebreaker is based on the Router ID (RID), which is a unique identifier for each OSPF router.
- The router with the highest RID will be selected as DR, and the second highest as BDR.
- The RID is usually the highest IP address configured on the router, or it can be set manually.

Election Process:

1. Initial Election:

- When OSPF routers on a broadcast network are connected, each router sends Hello messages to discover other routers on the same network.
- During this process, each router announces its priority and Router ID.
- Considering any router, if there is no other router with a higher priority (or higher Router ID in case of a tie), the router in question assumes the role of DR. If it finds a router with a higher priority, it considers that router as the DR and assumes the role of backup router (BDR) or simply as an adjacent router.

2. Determining the DR and BDR:

- The router with the highest priority and RID is elected as DR, and the second highest as BDR.
- If the DR or BDR fails, the BDR assumes the role of DR, and a new election for BDR is held.

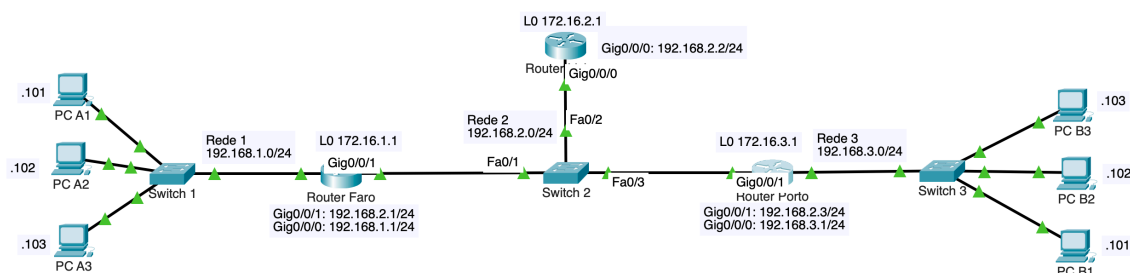
3. Subsequent Changes:

- Once the DR and BDR are elected, these roles do not change until one of the routers fails or the network is rebooted. Even if a new router with a higher priority is added to the network, it will not replace the current DR or BDR until a new election occurs.

Multicast addresses:

- 224.0.0.5 (AllSPFRouters): Used by non-DR and non-BDR routers to send packets to the DR and BDR.
- 224.0.0.6 (AllDRouters): Used by all routers, including the DR and BDR, to send packets to all OSPF routers on the network.

1 – Base topology to be implemented in Packet Tracer



The proposed architecture can be downloaded from the following link, where some of the basic configurations for the IP addressing of the physical interfaces have already been made, but must be confirmed.

In addition to the physical interfaces, loopback interfaces are also to be configured on each router. For this purpose, the following addresses must be assigned:

- Router Faro: **172.16.1.1**
- Router Lisboa: **172.16.2.1**
- Router Porto: **172.16.3.1**

The use of loopback interfaces has the advantage that they never fail, unlike physical interfaces. This stability ensures that the RID value of each router will never change, since, if not manually defined, it is determined by the highest IP address of the configured loopback interfaces and, if these do not exist, by the highest IP address of the physical interfaces. If the loopback interface is not active, if a physical interface becomes inactive, a new DR and BDR election process must take place.

2 – OSPF Configuration

The OSPF configuration is performed on each router, following the procedure already detailed in the previous practical lesson. The necessary steps are detailed below. Consider in this exercise that all routers are configured with the OSPF 100 process.

Faro router configuration

Suggested implementation, including both physical interfaces and loopback interfaces in the OSPF process.

```
Router> en
Router# conf t
Router-Faro (config)# int loopback 0
Router-Faro (config-if)# ip address 172.16.1.1 255.255.255.0
Router-Faro (config-if)# no shut down
Router-Faro (config-if)# exit
Router (config)# router ospf 100
Router-Faro (config-router)# network 192.168.1.0 0.0.0.255 area 0
Router-Faro (config-router)# network 192.168.2.1 0.0.0.0 area 0
Router-Faro (config-router)# network 172.16.1.1 0.0.0.0 area 0
Router-Faro (config-router)# exit
```

Lisboa router configuration

In this case, the interfaces covered by OSPF are only those referring to the IP address 192.168.2.2.

```
Router> en
Router# conf t
Router- Lisboa (config)# int loopback 0
Router- Lisboa (config-if)# ip address 172.16.2.1 255.255.255.0
Router- Lisboa (config-if)# no shut down
Router- Lisboa (config-if)# exit
Router (config)# router ospf 100
Router-Lisboa (config-router)# network 192.168.2.2 0.0.0.0 area 0
Router-Faro (config-router)# network 172.16.2.1 0.0.0.0 area 0
Router- Lisboa (config-router)# exit
```

Porto router configuration

```
Router> en
Router# conf t
Router-Porto (config)# int loopback 0
Router-Porto (config-if)# ip address 172.16.3.1 255.255.255.0
Router-Porto (config-if)# no shut down
Router-Porto (config-if)# exit
Router-Porto (config)# router ospf 100
```

```
Router-Porto (config-router)# network 192.168.3.0 0.0.0.255 area 0
Router-Porto (config-router)# network 192.168.2.3 0.0.0.0 area 0
Router-Faro (config-router)# network 172.16.3.1 0.0.0.0 area 0
Router-Porto (config-router)# exit
```

3 – Verification of routing tables

Using the show ip route command, verify that the routing tables are correctly populated and that all routers have routes to reach any destination on the network.

Router-Faro# show ip route:

```
Router-Faro#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/32 is subnetted, 3 subnets
C       172.16.1.1/32 is directly connected, Loopback0
O       172.16.2.1/32 [110/2] via 192.168.2.2, 00:11:36, GigabitEthernet0/0/1
O       172.16.3.1/32 [110/2] via 192.168.2.3, 07:07:26, GigabitEthernet0/0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.2.1/32 is directly connected, GigabitEthernet0/0/1
O       192.168.3.0/24 [110/2] via 192.168.2.3, 07:05:52, GigabitEthernet0/0/1
```

OSPF routes should be visible to the loopback interfaces of the other routers as well as to network 3, which is not directly connected. Note the metric of 2, indicating 2 hops through Ethernet networks.

Router-Lisboa# show ip route:

```
Router-Lisboa#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/32 is subnetted, 3 subnets
O       172.16.1.1/32 [110/2] via 192.168.2.1, 00:12:05, GigabitEthernet0/0/0
C       172.16.2.1/32 is directly connected, Loopback0
O       172.16.3.1/32 [110/2] via 192.168.2.3, 00:12:05, GigabitEthernet0/0/0
O       192.168.1.0/24 [110/2] via 192.168.2.1, 00:12:05, GigabitEthernet0/0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.2.2/32 is directly connected, GigabitEthernet0/0/0
O       192.168.3.0/24 [110/2] via 192.168.2.3, 00:12:05, GigabitEthernet0/0/0
```

In this case, the forwarding table should also contain an OSPF route for network 3. It should be noted that although there is only one router distance, the cost metric is 2. This is due to the

existence of a multi-access network segment between the routers, which causes both to contribute the costs of their interfaces to the total OSPF cost. In a P2P connection, the cost would only be 1.

Router-Porto# show ip route:

```
Router-Porto#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/32 is subnetted, 3 subnets
O       172.16.1.1/32 [110/2] via 192.168.2.1, 00:12:47, GigabitEthernet0/0/1
O       172.16.2.1/32 [110/2] via 192.168.2.2, 00:12:47, GigabitEthernet0/0/1
C       172.16.3.1/32 is directly connected, Loopback0
O       192.168.1.0/24 [110/2] via 192.168.2.1, 00:12:47, GigabitEthernet0/0/1
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.2.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.2.3/32 is directly connected, GigabitEthernet0/0/1
    192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.3.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.3.1/32 is directly connected, GigabitEthernet0/0/0
```

In this case, the expected behavior is similar, with a cost of reaching network 1, which is not directly connected to the Porto router.

4 – OSPF database verification

Using the command “show ip ospf database” you can confirm the LSDB on each router.

Router-Faro# show ip ospf database:

```
Router-Faro#show ip ospf d
Router-Faro#show ip ospf database
      OSPF Router with ID (172.16.1.1) (Process ID 100)

      Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum Link count
172.16.3.1   172.16.3.1     450      0x80000021   0x004443 3
172.16.1.1   172.16.1.1     445      0x8000001e   0x00e5ae 3
172.16.2.1   172.16.2.1     445      0x80000017   0x00b959 2

      Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
192.168.2.3  172.16.3.1     445      0x80000003   0x0018cf
192.168.2.2  172.16.2.1     445      0x80000002   0x0027c3
```

Router-Lisboa# show ip ospf database:

```
Router-Lisboa#sh ip ospf da
Router-Lisboa#sh ip ospf database
      OSPF Router with ID (172.16.2.1) (Process ID 100)

      Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum Link count
172.16.3.1   172.16.3.1     970      0x80000021   0x004443 3
172.16.2.1   172.16.2.1     966      0x80000017   0x00b959 2
172.16.1.1   172.16.1.1     966      0x8000001e   0x00e5ae 3

      Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
192.168.2.3  172.16.3.1     961      0x80000003   0x0018cf
192.168.2.1  172.16.1.1     968      0x80000012   0x0010cc
```

Router-Porto# show ip ospf database:

```
Router-Porto#sh ip o
Router-Porto#sh ip ospf d
Router-Porto#sh ip ospf database
      OSPF Router with ID (172.16.3.1) (Process ID 100)

      Router Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum Link count
172.16.3.1   172.16.3.1     582      0x80000021   0x004443 3
172.16.2.1   172.16.2.1     577      0x80000017   0x00b959 2
172.16.1.1   172.16.1.1     577      0x8000001e   0x00e5ae 3

      Net Link States (Area 0)

Link ID      ADV Router    Age      Seq#          Checksum
192.168.2.3  172.16.3.1     582      0x80000003   0x0018cf
192.168.2.2  172.16.2.1     572      0x80000002   0x0027c3
```

Looking at these tables, we can draw the following conclusions about the topology.

1. OSPF Router with ID (172.16.1) (Process ID 100)

Indicates the router ID and the associated OSPF process

2. Router Link States (Area 0)

The content is the Type 1 LSAs for each router in area 0 (the only one in this topology). Type 1 LSAs describe the router and its connections in the OSPF area.

Field	Description
Link ID	The Router ID of the router that originated the LSA.
ADV Router	The Router ID of the advertising router (which is the same as the Link ID in this case).
Age	The time in seconds since the LSA was generated or refreshed.
Seq#	The sequence number of the LSA, used to keep track of LSA updates. Higher numbers are newer.
Checksum	Used to verify the integrity of the LSA.
Link count	Number of links (interfaces) that this router has in the area.

It should be noted that, for example, the Faro and Porto routers have three connections in area 0. These are one connection to the other two routers and one connection to the multi-segment network.

3. Net Link States (Area 0)

The content is type 2 LSAs in area 0. Type 2 LSAs describe networks that include routers involved in the OSPF process.

Field	Description
Link ID	The IP address of the network.
ADV Router	The Router ID of the designated router (DR) that is responsible for this network segment.
Age	The time in seconds since the LSA was generated or refreshed.
Seq#	The sequence number of the LSA, used to keep track of LSA updates.
Checksum	Used to verify the integrity of the LSA.

Type 2 LSAs describe all multi-segment network segments existing in this topology. Thus, we find a total of three distinct type 2 LSAs: one for network 1, another for network 2, and another for network 3.

5 – Verification of neighborhood and adjacent relationships

Example for the Faro Router:

```
Router-Faro#show ip ospf n
Router-Faro#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.16.2.1	1	FULL/BDR	00:00:37	192.168.2.2	GigabitEthernet0/0/0
172.16.3.1	1	FULL/DR	00:00:37	192.168.2.3	GigabitEthernet0/0/0

As can be seen, the neighbors of the Faro router are the Lisbon and Porto routers, both with the same priority. You can also see that the relationships are already in FULL status, with the Porto router being the DR because it has the highest Router ID and the Lisbon router being the BDR because it has the second highest router ID. The Faro router is a DROTHER, indicating that it is neither DR nor BDR, as it has the lowest router ID.

Output for the Lisbon router:

```
Router-Lisboa>
Router-Lisboa>en
Router-Lisboa#sh ip osp
Router-Lisboa#sh ip ospf n
Router-Lisboa#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.16.1.1	1	FULL/DROTHER	00:00:35	192.168.2.1	GigabitEthernet0/0/0
172.16.3.1	1	FULL/DR	00:00:35	192.168.2.3	GigabitEthernet0/0/0

Viewing the neighbors of the Lisbon router confirms that the DR is the Porto router.

Output for the Porto router:

```
Router-Porto>
Router-Porto>en
Router-Porto#sh ip osp
Router-Porto#sh ip ospf n
Router-Porto#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.16.1.1	1	FULL/DROTHER	00:00:36	192.168.2.1	GigabitEthernet0/0/1
172.16.2.1	1	FULL/BDR	00:00:36	192.168.2.2	GigabitEthernet0/0/1

The Porto router has the Faro router and the Lisbon router as neighbors, as expected.

6 – Verification of connectivity from PCs

Example for connectivity between machine A1 and machine B3

C:\> **ping 192.168.3.103**

```
C:\>ping 192.168.3.103

Pinging 192.168.3.103 with 32 bytes of data:

Reply from 192.168.3.103: bytes=32 time<1ms TTL=126
Reply from 192.168.3.103: bytes=32 time=2ms TTL=126
Reply from 192.168.3.103: bytes=32 time<1ms TTL=126
Reply from 192.168.3.103: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.3.103:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
```

As expected, there is end-to-end connectivity.

7 – Reelection of a new DR

In this exercise, we want to force the Lisbon router to be the DR. To do this, we will start by confirming the current priority of this router using the command:

Router-Lisboa# show ip ospf interface GigabitEthernet0/0/0

```
Router-Lisboa#sh ip ospf interface g0/0/1
%OSPF: OSPF not enabled on GigabitEthernet0/0/1
Router-Lisboa#sh ip ospf interface g0/0/0

GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.2.2/24, Area 0
  Process ID 100, Router ID 172.16.2.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 172.16.3.1, Interface address 192.168.2.3
  Backup Designated Router (ID) 172.16.2.1, Interface address 192.168.2.2
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 2, Adjacent neighbor count is 2
    Adjacent with neighbor 172.16.1.1
    Adjacent with neighbor 172.16.3.1 (Designated Router)
  Suppress hello for 0 neighbor(s)
```

As you can see, this router, like all the others, has priority 1. So, let's increase the priority of this router:

Router-Lisboa# conf t

Router-Lisboa (config)# int g/0/0/0

Router-Lisboa (config-router)# ip ospf priority 255

Then, to trigger a new election, the following command shall be used:

Router-Lisboa (config-router)# **clear ip ospf process**

As you can see, the new DR is now the Lisbon router.

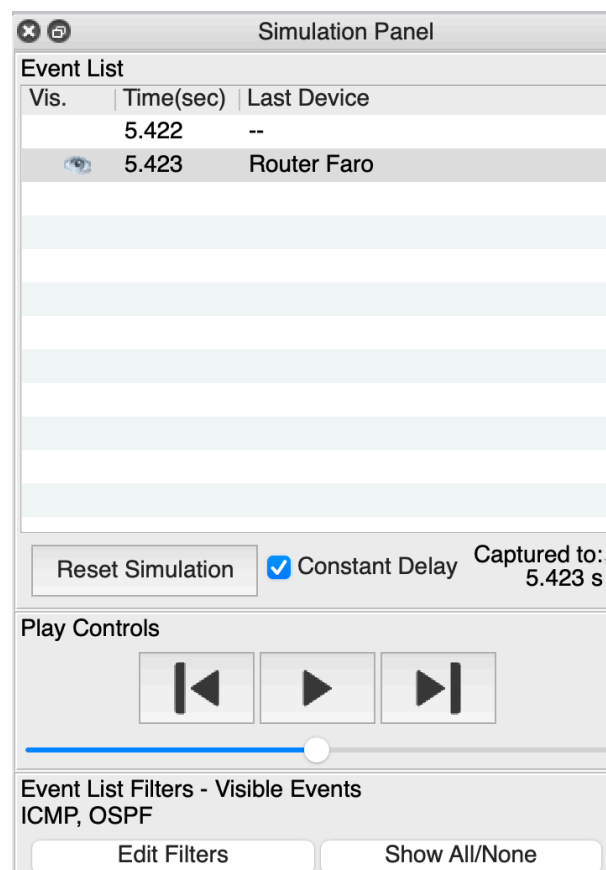
```
Router-Lisboa#sh ip ospf int
Router-Lisboa#sh ip ospf interface g0/0/0

GigabitEthernet0/0/0 is up, line protocol is up
  Internet address is 192.168.2.2/24, Area 0
  Process ID 100, Router ID 172.16.2.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 255
  Designated Router (ID) 172.16.2.1, Interface address 192.168.2.2
  Backup Designated Router (ID) 172.16.3.1, Interface address 192.168.2.3
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:00
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 2, Adjacent neighbor count is 2
    Adjacent with neighbor 172.16.3.1 (Backup Designated Router)
    Adjacent with neighbor 172.16.1.1
  Suppress hello for 0 neighbor(s)
```

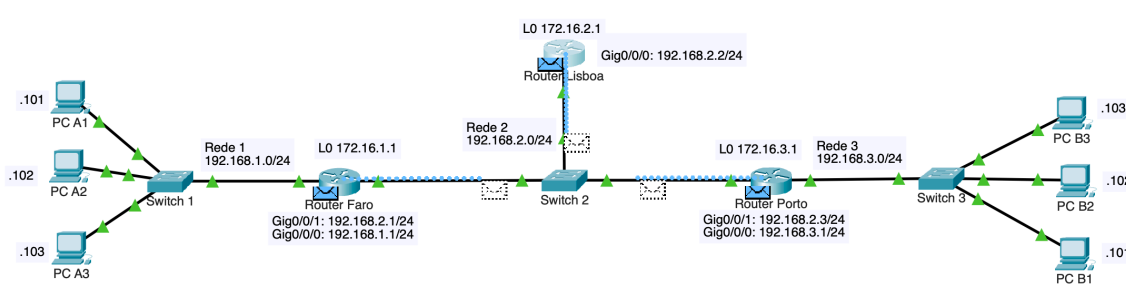
Note that in this case, the “Network Type” is Broadcast, denoting the type of Ethernet network.

8 – Viewing Messages in Simulation Mode

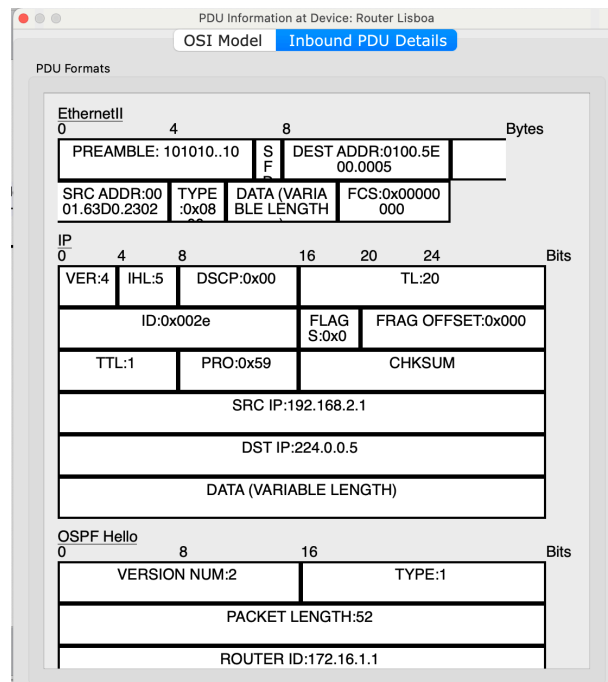
In this exercise, ensure that only ICMP and OSPF packets are filtered.



Use simulation mode in Packet Tracer to view packets with OSPF messages.



For example, the following message was sent by the Faro router to the Lisbon router (DR), so, as expected, the destination address is 224.0.0.5.



If the priority of one of the routers is changed again, for example, the Lisbon router to zero, we can see that messages during the phase in which the DR and BDR election is not yet complete are now sent to destination 224.0.0.6.

```
Router-Faro>
Router-Faro>en
Router-Faro#sh ip p
Router-Faro#sh ip osp
Router-Faro#sh ip ospf n
Router-Faro#sh ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
172.16.2.1	0	2WAY/DROTHER	00:00:32	192.168.2.2	GigabitEthernet0/0/1
172.16.3.1	1	FULL/DR	00:00:32	192.168.2.3	GigabitEthernet0/0/1

